# Global Olympic Sports Performance Analysis – Project Report

Oct 28, 2026 · @Vaibhav

## 1. Introduction

### 1.1 Project Overview

This project analyses Olympic medal performance across the Summer and Winter Games from 1992 to 2026, using four CSV datasets and Tableau Public. It covers 60 countries, 18 Games editions and 8,000 athlete records, and turns them into 7 charts, 4 KPI tiles, 2 dashboards and 1 story. The analysis asks which sport categories give the best medal return per athlete, which countries and regions are most efficient, how medal totals change over time, and whether home advantage and sports investment show up in the results.

| Dataset | Rows | Columns | What it holds |
|---|---|---|---|
| medal_results.csv | 910 | 19 | One record per country per Games: region, GDP tier, sports investment level, athletes sent, gold/silver/bronze, medal efficiency %, top sport, host-nation flag |
| games_summary.csv | 18 | 10 | One record per Games: host city and country, season, total athletes, participating nations, medals awarded, top country |
| athletes.csv | 8,000 | 26 | Athlete profile and career: country, sport, sport category, medals won, games participated, training hours, age, body metrics |
| alltime_country_rankings.csv | 60 | 13 | All-time totals per country: gold/silver/bronze, total medals, summer and winter medals, average efficiency %, overall rank |

### 1.2 Purpose

The purpose is to give sports organisations, funders and media a clear, data-backed view of Olympic success so they can make decisions instead of guessing. It is built around three users:

- **A National Olympic Committee strategist** who must decide which sport categories to invest in, and compare the home country with top performers.
- **A sports investment analyst** at a funding agency who needs to see whether higher investment produces more medals, and which regions are most efficient.
- **A broadcast data journalist** who needs year-by-year trends, host-nation stories and regional narratives for a documentary series.

## 2. Ideation Phase

### 2.1 Problem Statement

Olympic success depends on infrastructure, athlete development and investment, but 34 years of medal data across 60 countries and about 50 sports is too large to read by hand. Decision-makers cannot easily see which sports give the best return, which regions are efficient, or whether investment actually pays off.

| Who | Problem | Why it matters |
|---|---|---|
| National Olympic Committee strategist | Cannot tell which sport categories give the most medals per athlete | Wrong investment choices waste a four-year cycle |
| Sports investment analyst | Cannot link investment level to medal results by country and region | Grants may go to low-return programmes |
| Broadcast data journalist | Has no quick view of medal trends, host-nation effects and regional shifts | Stories lack evidence and context |

### 2.2 Empathy Map Canvas

The canvas below is drawn for the main user, the National Olympic Committee performance strategist.

| Quadrant | What we found |
|---|---|
| Says | "Where should we invest before the next Games?" "How do we compare with the top countries?" |
| Thinks | Some sports probably give a better return, but there is no clear evidence. Rivals like the United States and China seem out of reach. |
| Does | Reads long medal tables, compares spreadsheets by hand, prepares investment proposals for the committee |
| Feels | Pressure to justify budgets, unsure which discipline to back, overwhelmed by the volume of data |
| Pains | Scattered data, no per-athlete view, no regional comparison, slow reporting |
| Gains | One dashboard showing top sport categories, top countries, trends and efficiency, ready to present |

## 2.3 Brainstorming

Ideas were collected for each question the users ask, then narrowed to what the four datasets can answer in Tableau.

| Idea | Data needed | Decision |
|---|---|---|
| Average medals per athlete by sport category | athletes.csv | Built (Chart 1) |
| Total medals by country, top 10 | alltime_country_rankings.csv | Built (Chart 2) |
| Medals awarded per Games over time | games_summary.csv | Built (Chart 3) |
| Sports investment vs medals, per country | medal_results.csv | Built (Chart 4), with an Investment Score created from High/Medium/Low |
| Medal efficiency by region | medal_results.csv | Built (Chart 5) |
| Medals won by host nations | medal_results.csv | Built (Chart 6) |

| Idea | Data needed | Decision |
|---|---|---|
| Medal distribution by region | medal_results.csv | Built (Chart 7) |
| KPI tiles for participation, entries, nations and efficiency | all four files | Built (4 tiles) |
| Map of medals by country | alltime_country_rankings.csv | Kept as future scope |
| Athlete age and training-hours analysis | athletes.csv | Kept as future scope |

# 3. Requirement Analysis

## 3.1 Customer Journey Map

The journey follows the strategist from a business question to a decision, and shows where the dashboards remove friction.

| Stage | What the user does | Pain today | What the solution gives |
|---|---|---|---|
| 1. Question | Needs to choose sports to fund for the next cycle | No evidence, only opinions | A clear set of questions: best sport categories, best countries, trends, efficiency |
| 2. Explore | Opens medal tables and athlete lists | Four files, thousands of rows | One related data model across all four CSVs |
| 3. Analyse | Compares countries, sports and regions | Manual sums and comparisons | Sorted bar charts, trend line, bubble chart, regional pie and donut |
| 4. Decide | Picks target sports and regions for investment | Unsure which numbers to trust | Labelled values, KPI tiles and a consistent colour scheme |
| 5. Share | Presents to the committee, funders or media | Slides built from scratch | Two dashboards and a story published on Tableau Public |

## 3.2 Solution Requirement

### Functional requirements

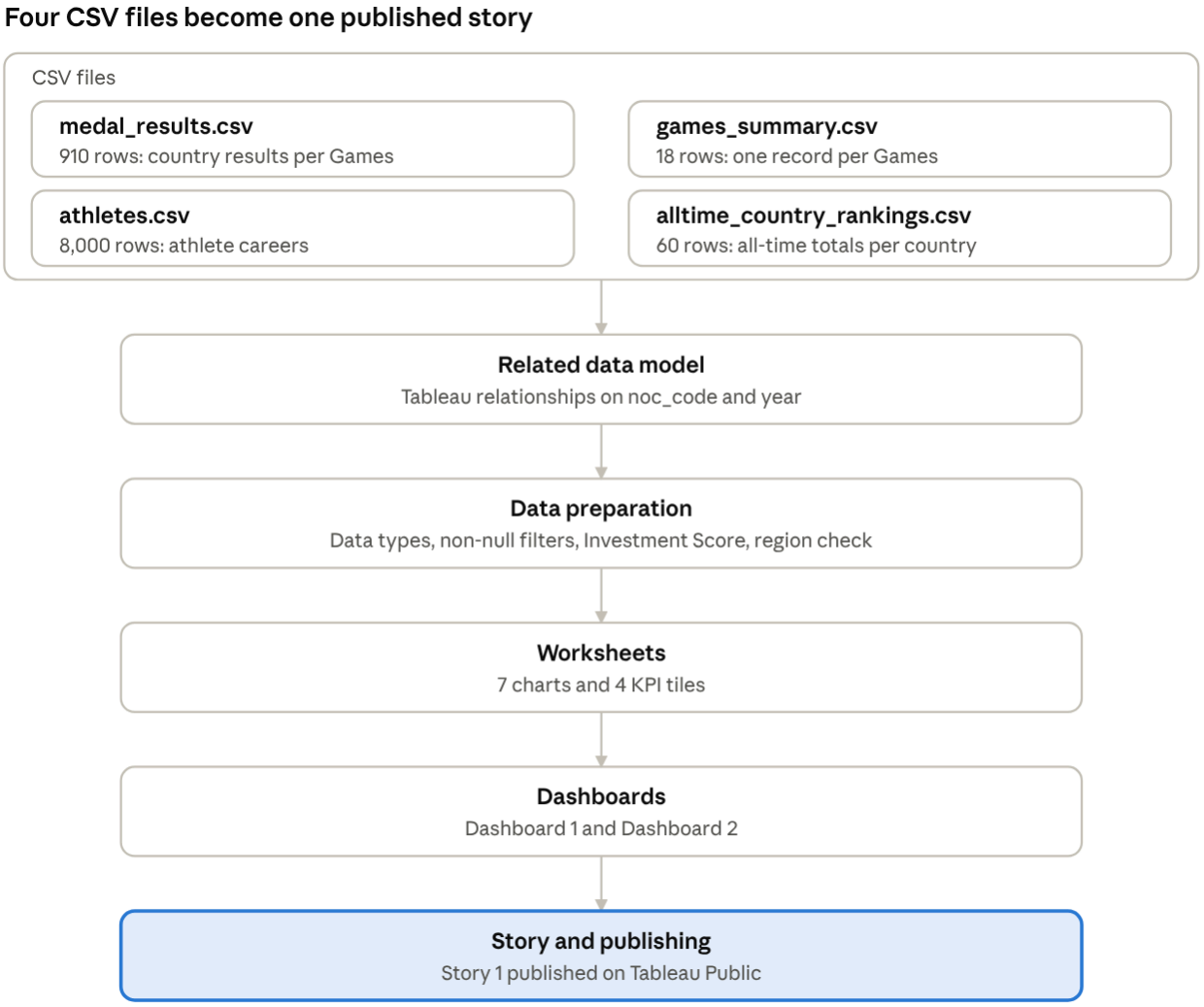
1. Load and relate four CSV datasets using shared keys (noc_code and year).
2. Clean and validate the data: correct data types, remove null medal records, confirm the seven region values.
3. Create calculated fields where the data needs it, such as an Investment Score from High / Medium / Low.
4. Build 7 charts and 4 KPI tiles that answer the three users' questions.
5. Combine them into two dashboards and one story with a clear order.
6. Publish the work to Tableau Public so it can be viewed and shared.

### Non-functional requirements

- Charts must be readable at a glance, with sorted bars, data labels and consistent colours.
- Totals must agree across datasets (see section 6).
- The workbook must run on a standard laptop and be viewable in any modern browser.

## 3.3 Data Flow Diagram

Data moves from the four CSV files into Tableau's related data model, is prepared, then drawn as sheets, dashboards and a story.




data flow · 4 CSV files to 1 story, 5 stages

Preparation happens once, in the data model, before any chart is built, so every sheet reads the same cleaned data.

## 3.4 Technology Stack

| Layer | Tool | Use in this project |
|---|---|---|
| Source data | 4 CSV files (Kaggle dataset from the project brief) | Medal results, Games summary, athletes, all-time country rankings |
| Data model | Tableau relationships | Joins the four files on noc_code and year without duplicating rows |

| Layer | Tool | Use in this project |
|---|---|---|
| Preparation | Tableau data source filters and calculated fields | Non-null medal filters, Investment Score (High = 3, Medium = 2, Low = 1) |
| Analysis and visuals | Tableau Public | 7 worksheets, 4 KPI sheets, 2 dashboards, 1 story |
| Publishing | Tableau Public (public.tableau.com) | Shareable, interactive version of the story |
| Environment | Windows PC with 8 GB RAM or more, any modern browser | As listed in the project's system requirements |

# 4. Project Design

## 4.1 Problem Solution Fit

Each problem from section 2.1 is matched to a visual that answers it directly.

| User problem | Solution built | Visual used |
|---|---|---|
| Which sport categories give the best medal return? | Average medals per athlete, by sport category, sorted high to low | Chart 1, horizontal bar |
| How does our country compare with the top performers? | Total medals of the top 10 countries | Chart 2, bar |
| How have medal totals moved over time? | Medals awarded per Games, 1992 to 2026 | Chart 3, line |
| Does more investment mean more medals? | Investment Score against total medals, sized by efficiency | Chart 4, bubble |
| Which regions convert athletes into medals best? | Average medal efficiency % by region | Chart 5, pie |
| Do host nations win more? | Total medals of host nations | Chart 6, bar |
| How are medals spread across regions? | Medal distribution by region | Chart 7, donut |
| What are the headline numbers? | Four KPI tiles | KPI sheets |

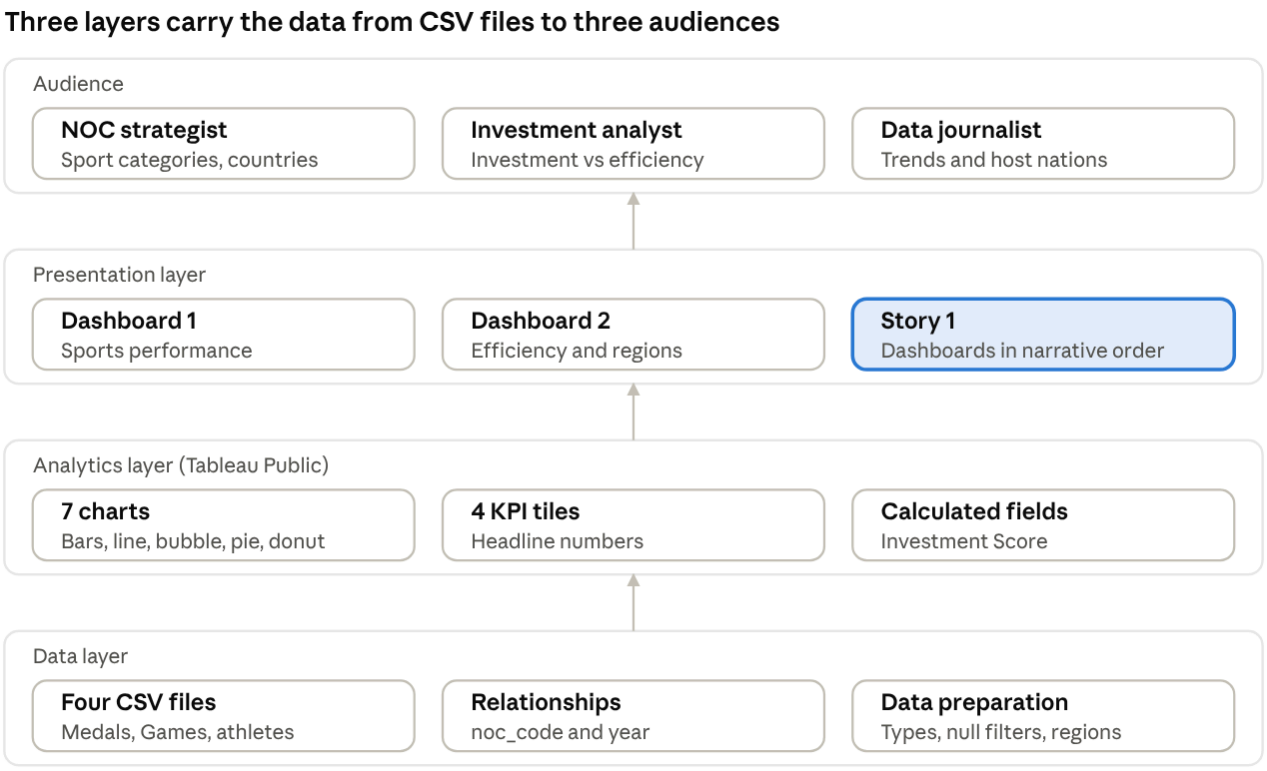
## 4.2 Proposed Solution

The solution is a two-dashboard Tableau Public workbook with a story on top. Dashboard 1 (Global Sports Performance and Analytics) covers sport categories, top countries and the medal trend. Dashboard 2 (Olympic Performance and Regional Efficiency) covers efficiency by region, investment against medals, host nations and the regional medal split. The story walks through both in a fixed order so the reader gets a narrative, not just charts.

| Component | Content | Count |
|---|---|---|
| Data model | Four CSVs related on noc_code and year | 1 |
| Calculated field | Investment Score from High / Medium / Low | 1 |
| Charts | Bar, line, bubble, pie, donut | 7 |
| KPI tiles | Athlete participation, categories entered, nations represented, regional efficiency | 4 |
| Dashboards | Performance and Analytics; Performance and Regional Efficiency | 2 |
| Story | Narrative over both dashboards | 1 |

## 4.3 Solution Architecture

The solution has three layers. Data is prepared at the bottom, analysed in Tableau in the middle, and presented as dashboards and a story to three audiences at the top.




solution architecture · 3 layers, 3 audiences

Arrows show the direction the data travels: nothing reaches an audience that has not passed through the analytics layer.

# 5. Project Planning and Scheduling

## 5.1 Project Planning

The project was run on SkillWallet as a series of epics, each with its own stories. The work took two days, 27 and 28 October 2026, and followed the order below. Each phase needed the previous one to be finished first.

| Phase (epic) | Main activities | Output | Dates | Status |
|---|---|---|---|---|
| 1. Data collection and preparation | Collect the dataset, connect the four CSVs in Tableau, relate them on noc_code and year | Related data model | 27 Oct | Completed |

| Phase (epic) | Main activities | Output | Dates | Status |
|---|---|---|---|---|
| 2. Data preparation | Check data types, add non-null filters, verify the seven regions, create the Investment Score | Clean, validated data | 27 Oct | Completed |
| 3. Data visualization | Build 7 charts and 4 KPI tiles, one worksheet each | 11 worksheets | 27 to 28 Oct | Completed |
| 4. Dashboard | Arrange the worksheets into two themed dashboards | Dashboard 1 and Dashboard 2 | 28 Oct | Completed |
| 5. Story | Order the dashboards into a narrative and publish to Tableau Public | Story 1 and public link | 28 Oct | Completed |
| 6. Performance testing | Cross-check totals, filters and rendering (section 6) | Test results | 28 Oct | Completed |
| 7. Report | Write this report with screenshots, GitHub link and dataset link | Project report | 28 Oct | In progress |

**Key dependencies.** The charts depend on the relationships being correct, because a wrong key would repeat rows and inflate every total. The dashboards depend on the worksheets keeping fixed names, and the story depends on both dashboards being final.

# 6. Functional and Performance Testing

Testing checked that the data is correct, the charts read it correctly and the dashboards render as designed. All checks below passed, and two defects found along the way were fixed.

## 6.1 Performance Testing

| # | Test | Method | Expected | Result |
|---|---|---|---|---|
| 1 | Record counts | Compared Tableau counts with the CSV files | 910, 18, 8,000 and 60 rows | Pass |

| # | Test | Method | Expected | Result |
|---|---|---|---|---|
| 2 | Numeric fields | Checked total_medals, gold, silver, bronze, medal_points, athletes_sent and efficiency fields | All numeric | Pass |
| 3 | Region values | Dragged Region onto Rows | Exactly 7 values | Pass |
| 4 | Null medal records | Added non-null filters on Total Medals, Gold, Silver, Bronze | No null medal values in charts | Pass |
| 5 | Investment Score | Plotted score against Sports Investment | High = 3, Medium = 2, Low = 1, no null | Pass |
| 6 | Totals across datasets | Summed medals awarded per Games (games_summary) and total medals per country (alltime_country_rankings) | Both give 6,025 | Pass |
| 7 | Top 10 filter | Filtered countries by sum of total medals | 10 bars, total 3,833 | Pass |
| 8 | Host-nation filter | Filtered medal_results to Host Nation = 1 | 13 host countries, total 426 medals | Pass |
| 9 | Dashboard rendering | Opened both dashboards | All charts and KPI tiles visible and readable | Pass |
| 10 | Published story | Opened the Tableau Public link | Story loads with both dashboards | Pass |

**Defects found and fixed**

| Defect | Cause | Fix |
|---|---|---|
| Chart 1 bar labels showed medal sums (431, 318, ...) while the bars showed averages | A SUM field was left on the Label shelf | Replaced it with the average, so labels read 1.6902, 1.6563, ... |
| Chart 2 listed all 60 countries | Top 10 filter not yet applied | Added the Top 10 filter by sum of total medals |

*[Optional screenshot: Data Source tab showing the relationships between the four tables]*

# 7. Results

## 7.1 Output Screenshots

The figures below show each output with the values read from it. All values come from the supplied CSV files. Paste the matching screenshot into each marked slot.

**Figure 1. Average medals per athlete by sport category**

Biathlon gives the best return per athlete, but the whole range is narrow: the top category is about 29% above the bottom one.

- Highest: Biathlon 1.6902, Shooting 1.6563, Cycling 1.6180, Track & Field 1.5667, Team Ball 1.5491
- Lowest: Gymnastics 1.3352 and Sliding 1.3098
- 21 sport categories, sorted high to low

*[Insert screenshot: Avg Medals per Athlete by Sport Category]*

**Figure 2. Global performance by country (top 10)**

The United States and China lead by a wide margin. Together they hold 1,431 medals, about 24% of all 6,025 medals.

- United States 735, China 696, Russia 429, Great Britain 401, Germany 336, Japan 308, Australia 280, Italy 198
- France and South Korea complete the top 10; the 10 countries total 3,833 medals

*[Insert screenshot: Global Performance Metrics by Country]*

**Figure 3. Trend of total medals awarded over the years**

Medals awarded per Games stay in a tight band, so the size of the Games is stable and differences between countries are not driven by a growing medal pool.

- Low: 305 medals in 2000. Peak: 363 medals in 2020. Latest: 334 in 2026
- 18 Games from 1992 to 2026, 6,025 medals in total, an average of about 335 per Games

*[Insert screenshot: Trend of Total Medals Awarded Over the Years]*

**Figure 4. Global sports investment vs medal performance (bubble chart)**

Higher investment goes with higher medal totals. Each bubble is one country, placed by its average Investment Score (Low = 1, Medium = 2, High = 3) and its total medals, and sized by average medal efficiency.

- The highest totals, up to about 735 medals, sit at score 3; countries near score 1 sit at the bottom of the chart

- Bubble size ranges from an efficiency of 9.11 to 108.30, so some small-total countries are highly efficient
- 60 countries, coloured by region

*[Insert screenshot: Global Sports Investment vs. Medal Performance]*

**Figure 5. Average medal efficiency % by region (pie chart)**

South America is the most efficient region, and the two regions that win the most medals in total are the least efficient per athlete.

| Region | Avg medal efficiency % |
|---|---|
| South America | 49.64 |
| Europe/Asia | 45.98 |
| Africa | 42.80 |
| Asia | 32.96 |
| Oceania | 27.79 |
| Europe | 27.48 |
| North America | 27.46 |

*[Insert screenshot: Avg Medal Efficiency % by Region]*

**Figure 6. Medal totals across Olympic hosts**

Host nations win a lot of medals, and two of them dominate: China (118) and the United States (116) account for 234 of the 426 host-nation medals, about 55%.

- Next: Japan 51, Russia 39, Italy 30, France 15, Australia 14, Norway 13, South Korea 11, Canada 10, Brazil 6, Greece 2, Spain 1
- 13 host countries, totals taken across the Games each one hosted

*[Insert screenshot: Comparing Medal Totals Across Olympic Hosts]*

**Figure 7. Regional medal distribution (donut chart)**

Medals are concentrated in a few regions: Europe takes the largest share, followed by Asia and North America. Europe/Asia, Africa and South America are the thinnest slices.

*[Insert screenshot: Regional Medal Efficiency and Distribution donut chart]*

**Figure 8. KPI tiles**

| KPI | Value | How it is calculated |
|---|---|---|
| Global Athlete Participation | 910 | Records in medal_results (country entries across all Games) |
| Total Sports Categories Entered | 60 | Unique NOC entries in alltime_country_rankings |
| Total Nations Represented | 18 | Records in games_summary (one per Games, 1992 to 2026) |
| Regional Performance Efficiency | 31.8 | Average medal efficiency % across the 910 records |

*[Insert screenshot: the four KPI tiles]*

**Figure 9. Dashboard 1: Global Sports Performance and Analytics**

Shows the three headline KPIs, average medals per athlete by sport category, the top countries and the medal trend on one screen, so the strategist can compare sports and countries together.

*[Insert screenshot: Dashboard 1]*

**Figure 10. Dashboard 2: Olympic Performance and Regional Efficiency**

Shows participation and efficiency KPIs, regional efficiency, investment against medals, host-nation medals and the regional split, which answers the funding analyst's questions in one view.

*[Insert screenshot: Dashboard 2]*

**Figure 11. Story 1**

The story presents both dashboards in order, from global performance to regional efficiency. It is published on Tableau Public: open the story.

*[Insert screenshot: Story 1]*

## 8. Advantages and Disadvantages

The workbook answers its three users' questions clearly, but it rests on static data and a few simplifications that readers should know about.

**Advantages**

- One related data model replaces four separate files, so every chart reads the same cleaned data.

- Each user gets a direct answer: sport categories (Figure 1), countries (Figure 2), investment against medals (Figure 4) and regional efficiency (Figure 5).
- Totals reconcile across datasets: both games_summary and alltime_country_rankings give 6,025 medals.
- Sorted bars, data labels and a single colour scheme make the charts readable at a glance.
- Publishing on Tableau Public makes the story shareable with a link, with no software needed to view it.
- The Investment Score turns a text field (High / Medium / Low) into a number that can be plotted, without changing the source file.

**Disadvantages**

- The data is a fixed set of CSV files, so the dashboards do not update when new results arrive.
- Sports investment is only High, Medium or Low, so the bubble chart shows a trend, not exact spending against medals.
- Medal efficiency is medals per athlete sent, which can favour small delegations: values go as high as 108.30%.
- Averages in Figure 1 and Figure 5 hide how much individual athletes and countries differ.
- Tableau Public workbooks are publicly visible, so the approach is unsuitable for confidential data.
- Two KPI tiles use broad labels: the value 60 counts countries, and 18 counts Games editions.

# 9. Conclusion

The analysis shows that Olympic success is concentrated, steady in size and linked to investment. It turns 34 years of data on 60 countries into two dashboards and a story that a strategist, a funder or a journalist can read in minutes.

- **Sport categories:** Biathlon (1.6902), Shooting (1.6563) and Cycling (1.6180) give the best medals per athlete, though the gap to the lowest category, Sliding (1.3098), is only about 29%.
- **Countries:** the United States (735) and China (696) lead by a wide margin and hold about 24% of all 6,025 medals.
- **Trend:** medals awarded per Games stay between 305 and 363, so the medal pool is stable.

- **Investment:** the highest medal totals sit at the highest investment score, supporting the case for sustained funding.
- **Regions:** South America (49.64%) and Europe/Asia (45.98%) are the most efficient per athlete, while Europe and North America win the most but convert least efficiently.
- **Hosts:** host nations won 426 medals in total, with China (118) and the United States (116) far ahead.

For the three users, this means a committee can target high-return categories, a funder can look beyond the biggest medal winners to efficient regions, and a journalist has evidence for trend and host-nation stories. The project also met its technical goals: four datasets were related and validated, 7 charts and 4 KPI tiles were built, and the results were published as two dashboards and a story.

## 10. Future Scope

The datasets hold more than the first version uses, and several extensions need no new data.

1. **Fairer efficiency measures.** Use population_million and gdp_tier from medal_results to show medals per million people and results by income tier.
2. **Map view.** Plot total medals by country on a filled map, using the country geographic role.
3. **Athlete analysis.** Use age, training hours per week and professional status in athletes.csv to see what separates medal winners.
4. **Real investment figures.** Replace the High / Medium / Low level with actual spending, so the bubble chart can show a true return on investment.
5. **More interactivity.** Add filters and dashboard actions for season (Summer or Winter), region and year range.
6. **Forecasting.** Add a forecast on the medal trend, and projected medals for the next Games.
7. **Live data and sharing.** Connect to a database or refreshed file and publish on Tableau Server or Tableau Cloud for organisations that cannot use public hosting.
8. **Integrity checks.** Study the doping_violation and world_record_holder fields alongside medal results.

# 11. Appendix

## Links

| Item | Link |
|---|---|
| Dataset link | *[Add dataset link here]* |
| GitHub link | github.com/abby0406/Global_Olympic_Sports_Analysis |
| Project demo link (Tableau Public story) | Open the story |

## Source Code

The project is a Tableau workbook, so there is no program code. The one calculated field, Investment Score, turns the text investment level in medal_results.csv into a number:

```
CASE UPPER(TRIM([Sports Investment]))
WHEN "HIGH" THEN 3
WHEN "MEDIUM" THEN 2
WHEN "MED" THEN 2
WHEN "LOW" THEN 1
END
```

## Data Source Relationships

| From | To | Key |
|---|---|---|
| medal_results | games_summary | year |
| medal_results | alltime_country_rankings | noc_code |
| medal_results | athletes | noc_code |

## Filters Applied

| Where | Filter |
|---|---|
| Data source | Non-null values on Total Medals, Gold, Silver and Bronze |
| Chart 2 | Top 10 countries by sum of total medals |

| Where | Filter |
| --- | --- |
| Chart 6 | Host Nation = 1 |